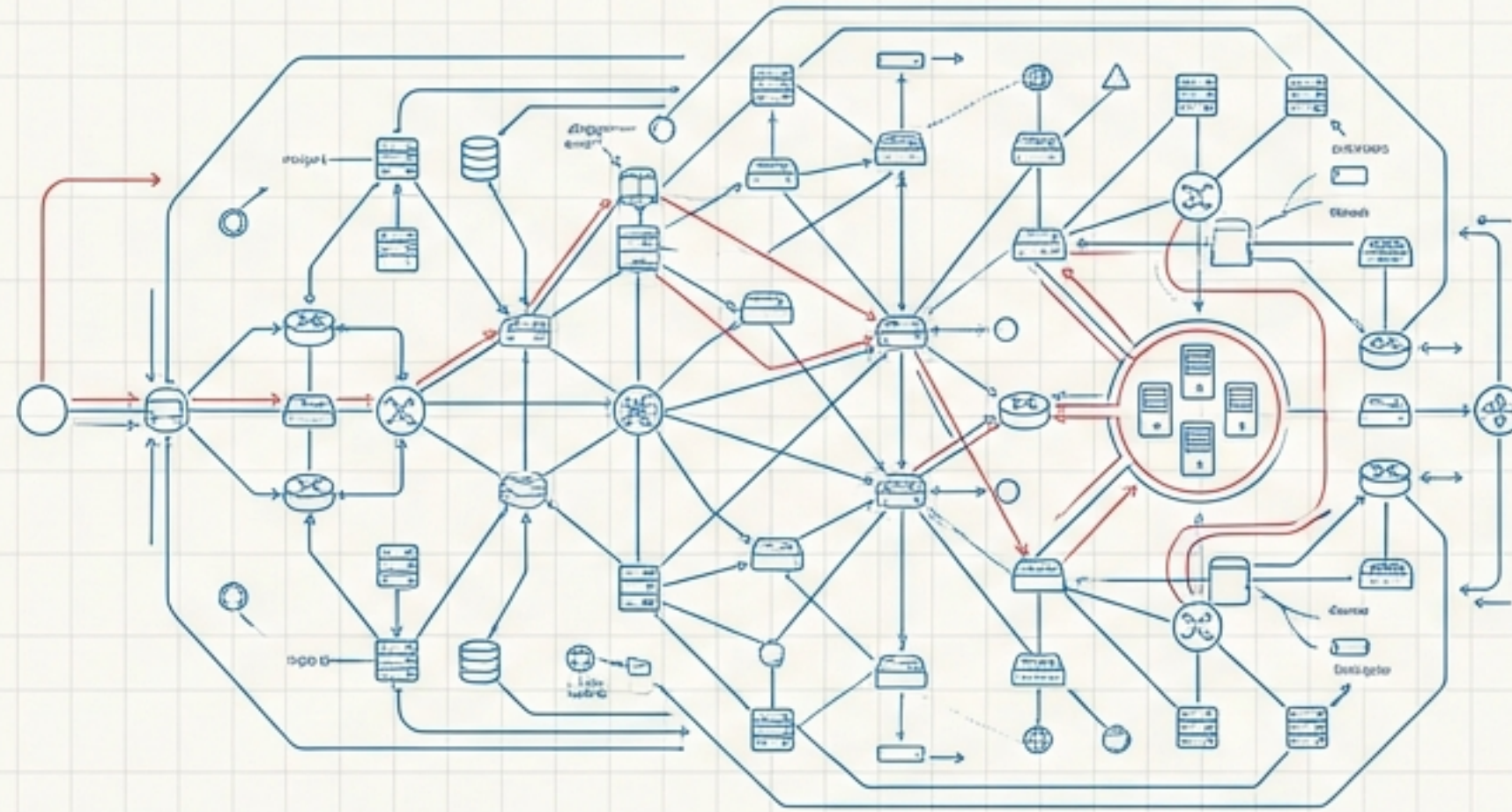


Every Cyberattack is a Network Event



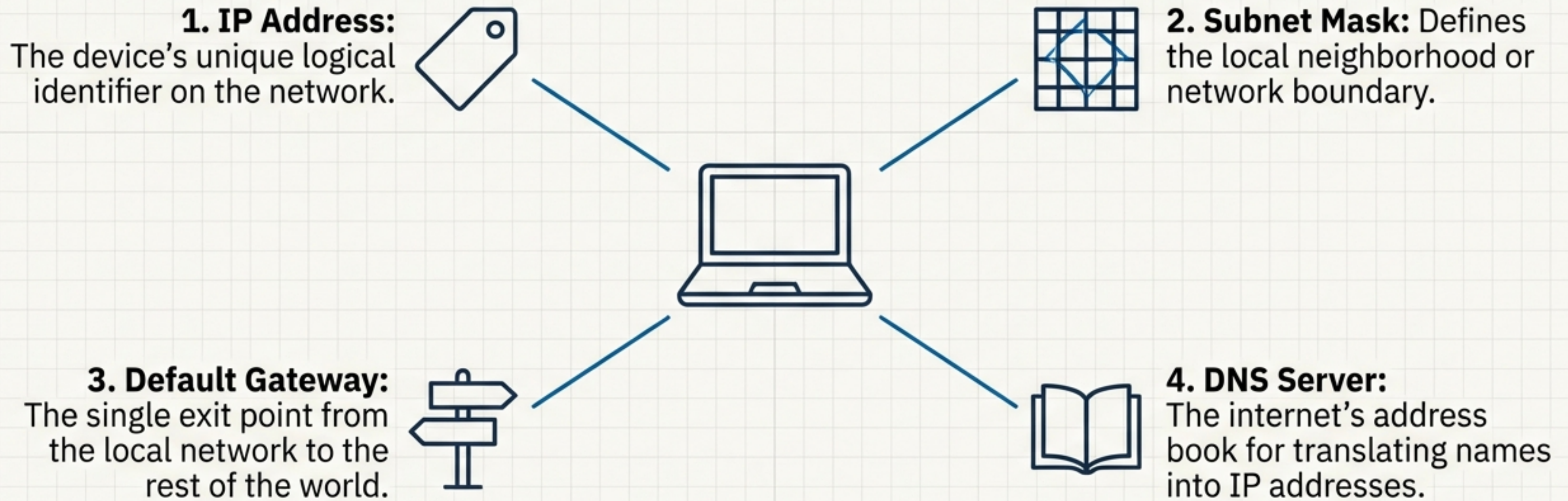
Before you can defend a territory, you must understand its layout, its pathways, and its language. In cybersecurity, that territory is the network.

Understanding how data flows is the foundation for detecting anomalies and responding to threats.

This session provides the blueprint. We will deconstruct the network to understand its components, its protocols, and its inherent weaknesses.

The Blueprint for Connection: 4 Essential Addresses

For any device to join the internet, four parameters must be configured. Together, they define its identity, boundary, route, and ability to find others.



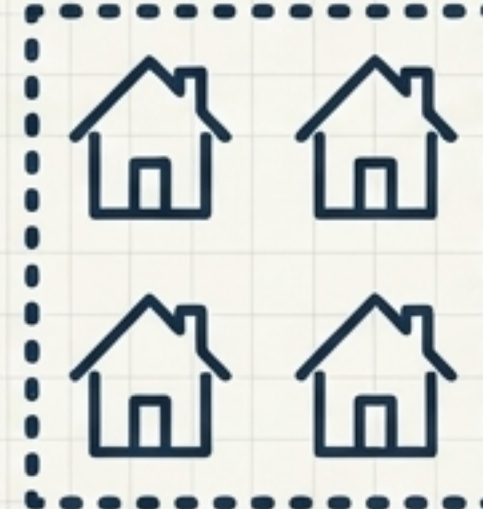
Identity and Boundaries: A Practical Analogy



IP Address (e.g., 192.168.1.10): Your Street Address

Uniquely identifies your house (device) on the street.

Supports IPv4 (32-bit) and IPv6 (128-bit) formats.



Subnet Mask (e.g., 255.255.255.0): Your Neighborhood Boundary

Defines which houses are local.
Determines if traffic stays on your street or needs to be sent out.



Security Note: Incorrect subnetting can break network segmentation, exposing critical systems.



Default Gateway (e.g., 192.168.1.1): The Road to the World

The router that forwards all traffic destined for outside your neighborhood.



Security Role: The first line of defense for filtering inbound/outbound traffic.



DNS Server: The Internet's Address Book

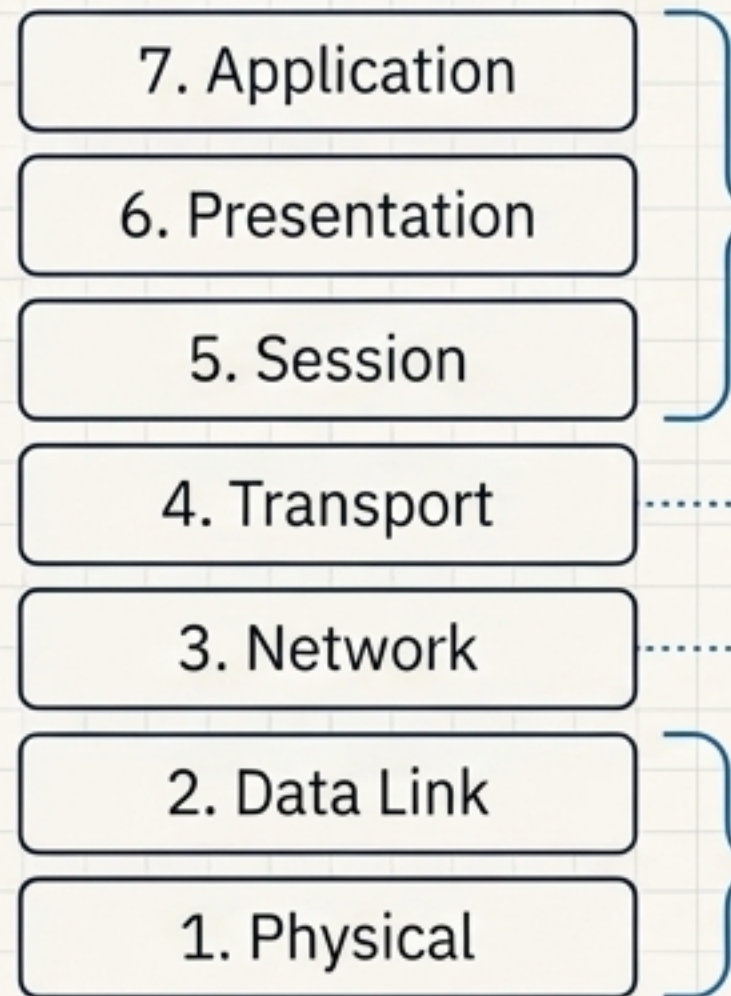
Translates a known name (www.wu.ac.th) into an address (203.158.xxx.xxx).



Security Threat: Vulnerable to DNS Spoofing, which can redirect users to malicious sites. Secure alternatives include DNSSEC and DNS over HTTPS (DoH).

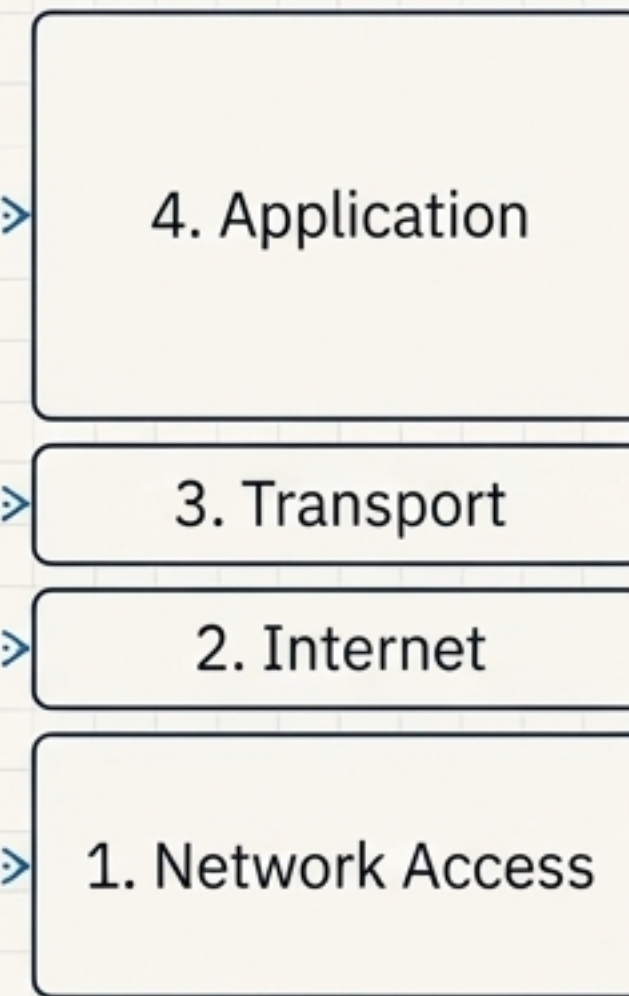
The Architectural Models: OSI vs. TCP/IP

OSI Model (The 7-Layer Reference)



A complete, conceptual framework used for teaching and design.

TCP/IP Model (The 4-Layer Practical Stack)



The model that powers the modern internet.

While the OSI model is the comprehensive theory, TCP/IP is the implemented reality. We will focus on the TCP/IP stack to understand real-world network traffic.

The Transport Layer: Choosing Between Reliability and Speed



TCP (Transmission Control Protocol)

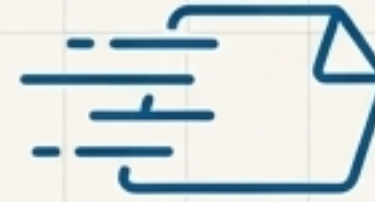
Purpose: Provides reliable, ordered, and error-checked data delivery.

Process: Uses a "Three-Way Handshake" (SYN, SYN/ACK, ACK) to establish a connection before sending data.

Used For: Web browsing (HTTPS), email (SMTP), file transfers (FTP).



Weakness: The handshake process can be exploited for SYN Flood (Denial of Service) attacks.



UDP (User Datagram Protocol)

Purpose: Provides fast, lightweight, connectionless data transmission.

Process: Sends data without establishing a connection or guaranteeing delivery. "Fire and forget."

Used For: DNS queries, video/audio streaming (VoIP), online gaming.



Weakness: Lack of connection state makes it easy to spoof source IPs, enabling amplification and flood attacks.

The Language of the Web: HTTP vs. HTTPS

HTTP (Hypertext Transfer Protocol)

Function

The foundational protocol for transferring data on the web. Operates on port 80.

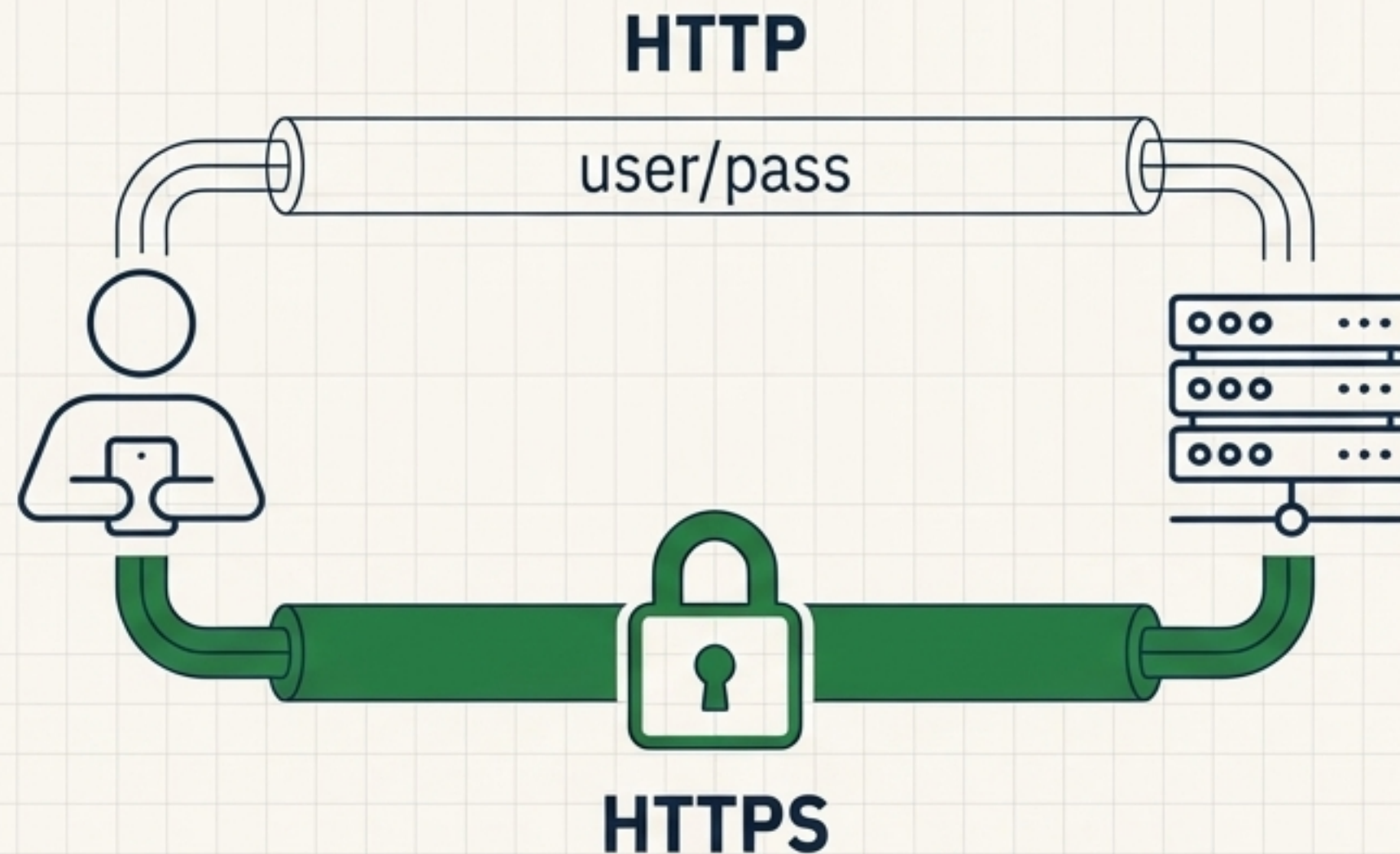
Characteristics

Transmits data in plain text, making it completely readable to anyone who can capture the packets.



Critical Weakness

Susceptible to eavesdropping (sniffing) and Man-in-the-Middle (MITM) attacks for credential theft.



HTTPS (HTTP Secure)

Function

HTTP layered over TLS/SSL encryption. Operates on port 443.

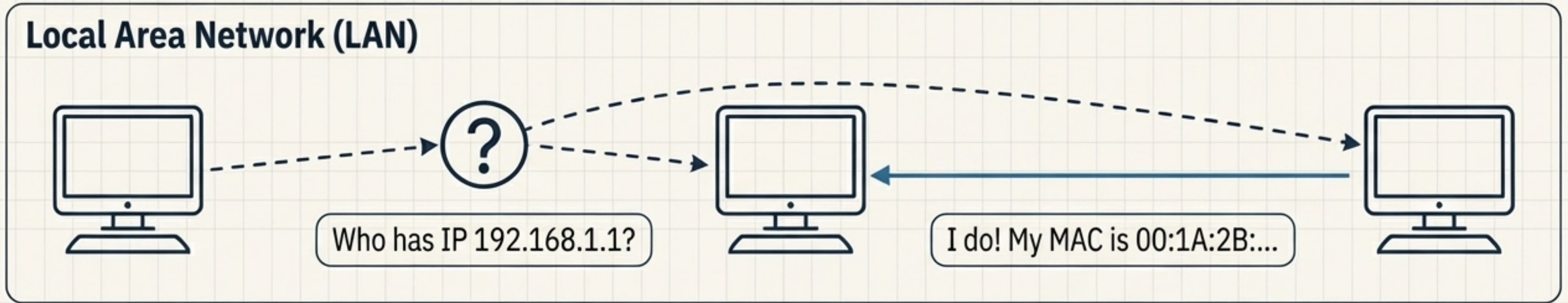
Core Benefits

1. **Encryption:** Protects data confidentiality.
2. **Authentication:** Verifies the server's identity via certificates.
3. **Integrity:** Ensures data has not been altered in transit.

Security Note

Still vulnerable if certificate management is poor, leading to phishing or downgrade attacks.

The Local Directory: Address Resolution Protocol (ARP)



Function

Maps a known Layer 3 (IP) address to its corresponding Layer 2 (MAC) address.

Analogy: Within a single room, asking "Who is John Smith?" and having that person raise their hand. It's essential for devices on the same local network to find each other.

The Inherent Weakness

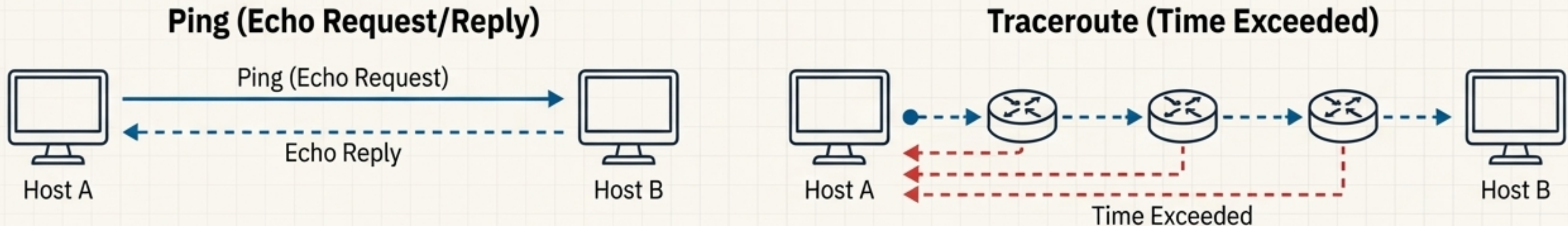
ARP operates on trust. There is **no authentication or verification** mechanism. Any device on the network can claim to have any IP address.

Primary Attack: ARP Spoofing (Poisoning)

An attacker sends forged ARP replies to associate their MAC address with the IP address of the default gateway or another victim.

Result: All traffic from the victim is routed through the attacker, enabling a **Man-in-the-Middle (MITM)** attack to intercept or modify data.

Network Diagnostics & Control: Internet Control Message Protocol (ICMP)



Legitimate Purpose

- Used by network devices to send error messages and operational information.
 - **Ping (Echo Request/Reply):** Tests host reachability.
 - **Traceroute (Time Exceeded):** Maps the path packets take to a destination.

The Double-Edged Sword

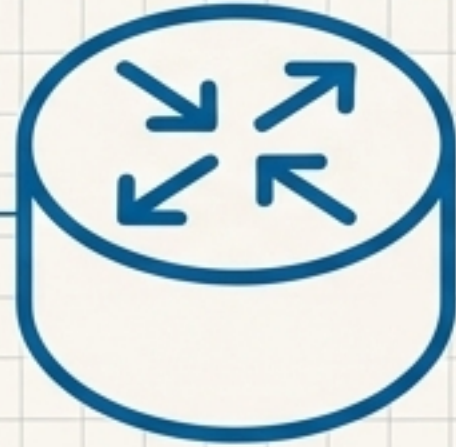
- 🛡️ **Reconnaissance:** Attackers use 'ping sweeps' to map out live hosts on a target network.
- 🛡️ **Denial of Service:** ICMP flood attacks (like the classic 'Smurf attack') can overwhelm a target with traffic.
- 🛡️ **Covert Channels:** ICMP packets can be manipulated to carry hidden data (ICMP tunneling), exfiltrating information past firewalls.

The Guardian Toolkit: Core Network Security Devices

Router

Function: Routes traffic between different networks.

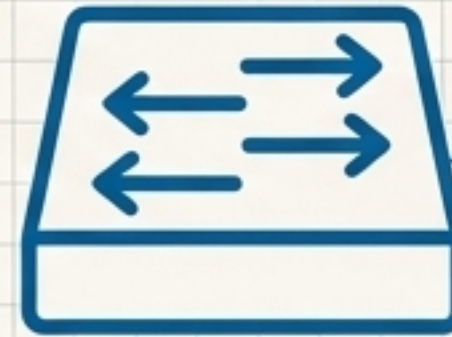
Security Role: Enforces network boundaries and controls traffic paths with Access Control Lists (ACLs).



Switch

Function: Connects devices within a single Local Area Network (LAN).

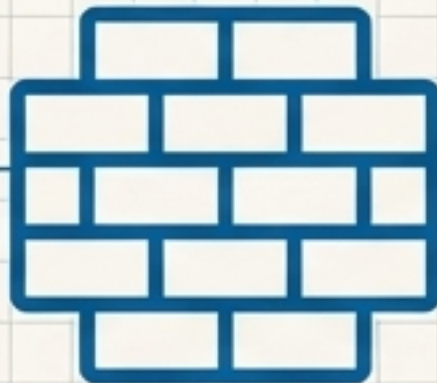
Security Role: Enables network segmentation through VLANs to isolate traffic and prevent snooping.



Firewall

Function: Filters traffic based on a predefined set of security rules.

Security Role: The primary gatekeeper, blocking malicious connections based on source/destination IP, port, and protocol.



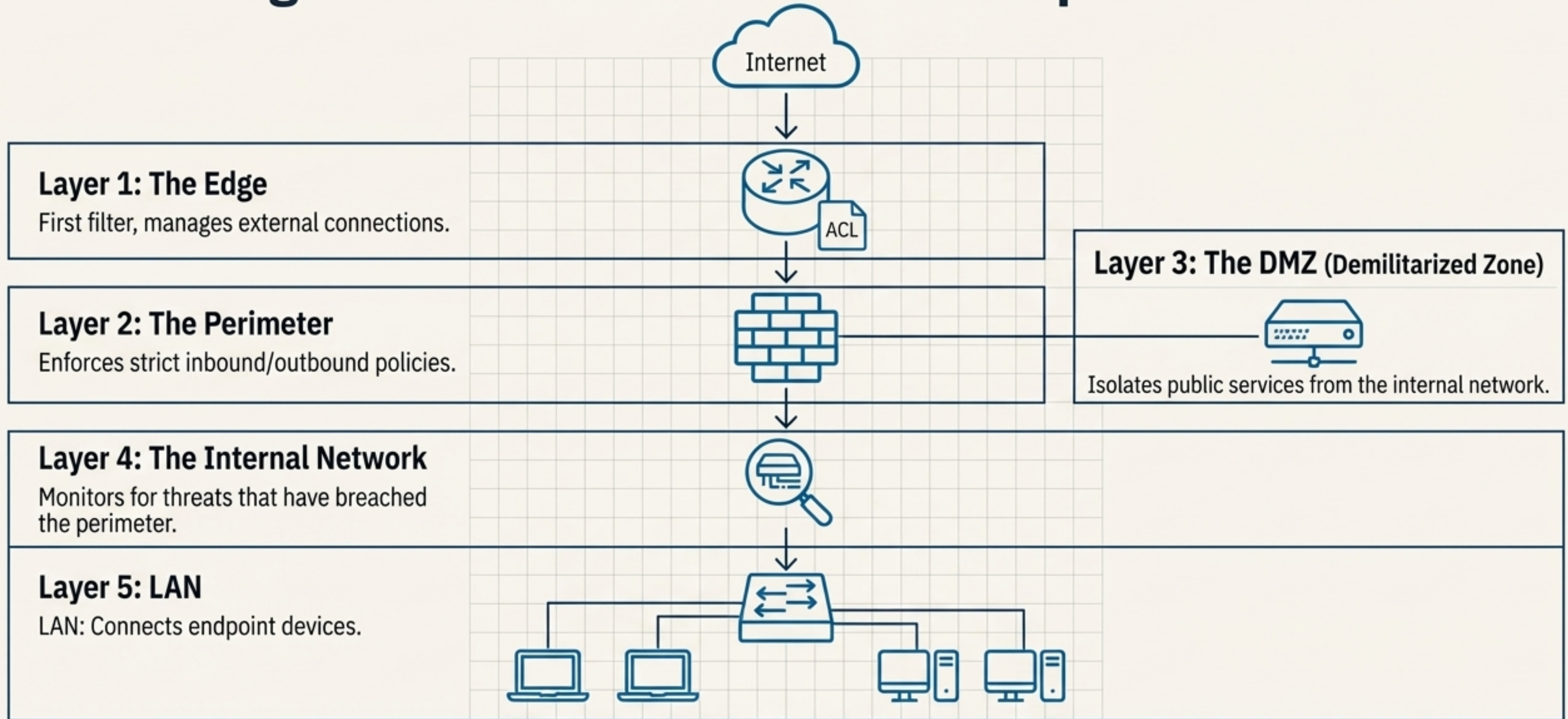
IDS/IPS (Intrusion Detection/Prevention System)

Function: Monitors network traffic for malicious activity or policy violations.

Security Role: The security guard; detects (IDS) or actively blocks (IPS) threats that get past the firewall.

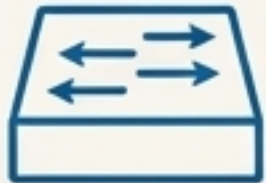
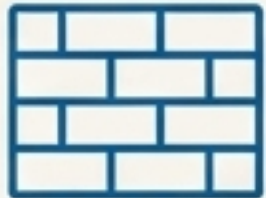


Assembling the Fortress: A Defense-in-Depth Architecture

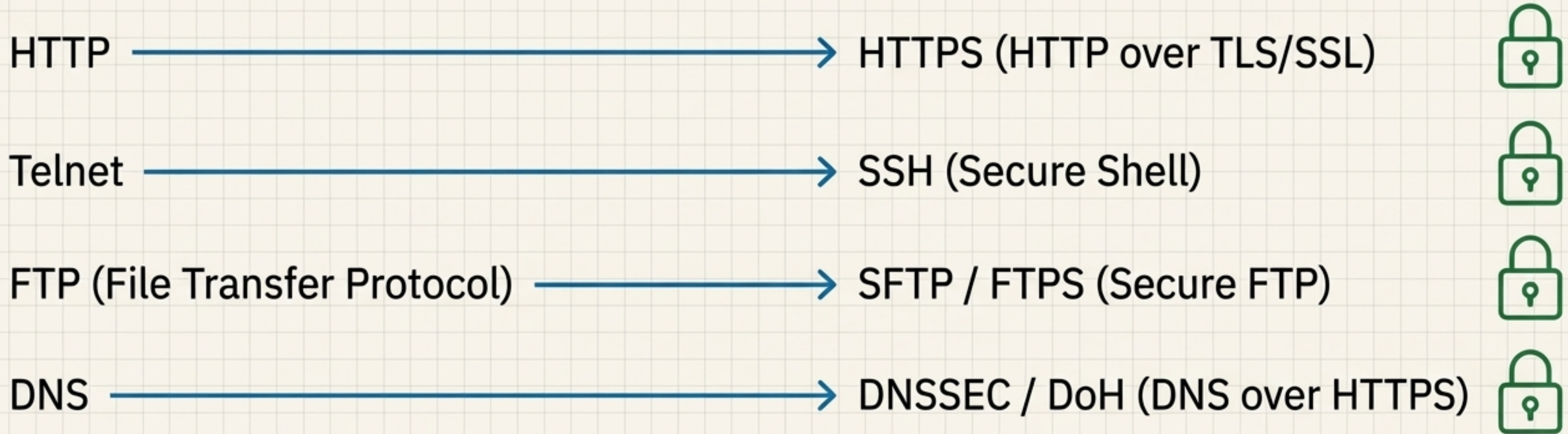


A secure architecture is not built on a single device. It is created by layering controls, ensuring that if one defense fails, others are in place to detect or stop the attack.

Device Weaknesses and Fortification Strategies

Device	Potential Weakness	Mitigation
 Router	Default credentials , outdated firmware, misconfigured ACLs.	Change all default passwords , implement a regular patching schedule, audit ACL rules.
 Switch	ARP spoofing , VLAN hopping attacks.	Implement port security (limit MAC addresses per port), properly configure trunk ports.
 Firewall	Overly permissive or misconfigured rules ("any/any").	Conduct regular rule base reviews , follow the principle of least privilege , log and audit changes.
 Access Point (AP)	Weak encryption (WEP/WPA), rogue APs, default admin panels.	Enforce WPA3 encryption , conduct wireless scans to detect rogue devices, change admin credentials.
 VPN Gateway	Weak encryption ciphers, leaked keys, configuration flaws.	Use strong , modern encryption algorithms; require Multi-Factor Authentication (MFA).

The Evolution of Security: From Plaintext to Encrypted



Core Lesson

The history of network protocol development is a history of adding layers of security in response to discovered weaknesses. Most legacy protocols were designed for functionality in a trusted environment, not for security in a hostile one.

Key Security Principles Introduced

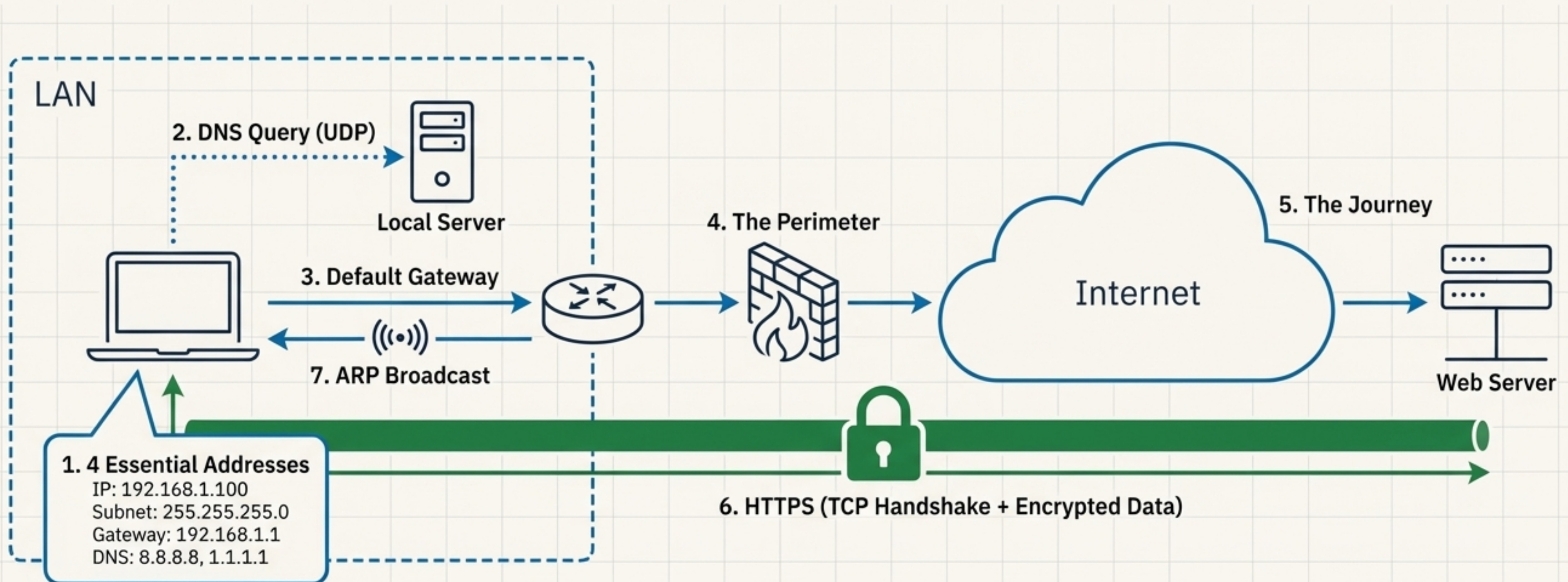
Encryption (confidentiality), Authentication (identity), and Validation (integrity).

A Map of Inherent Risks in the TCP/IP Stack

	Layer	Protocol(s)	Inherent Weakness	Common Attack Vector
Application	Application	HTTP, DNS, FTP	Unencrypted data, lack of verification	Credential Sniffing, DNS Cache Poisoning
Transport	Transport	TCP, UDP	Connection state abuse, no connection state	SYN Flood (DoS), UDP Amplification Attacks
Internet	Internet	IP, ICMP, ARP	No built-in authentication or identity verification	IP Spoofing, Ping Floods, ARP Poisoning (MITM)

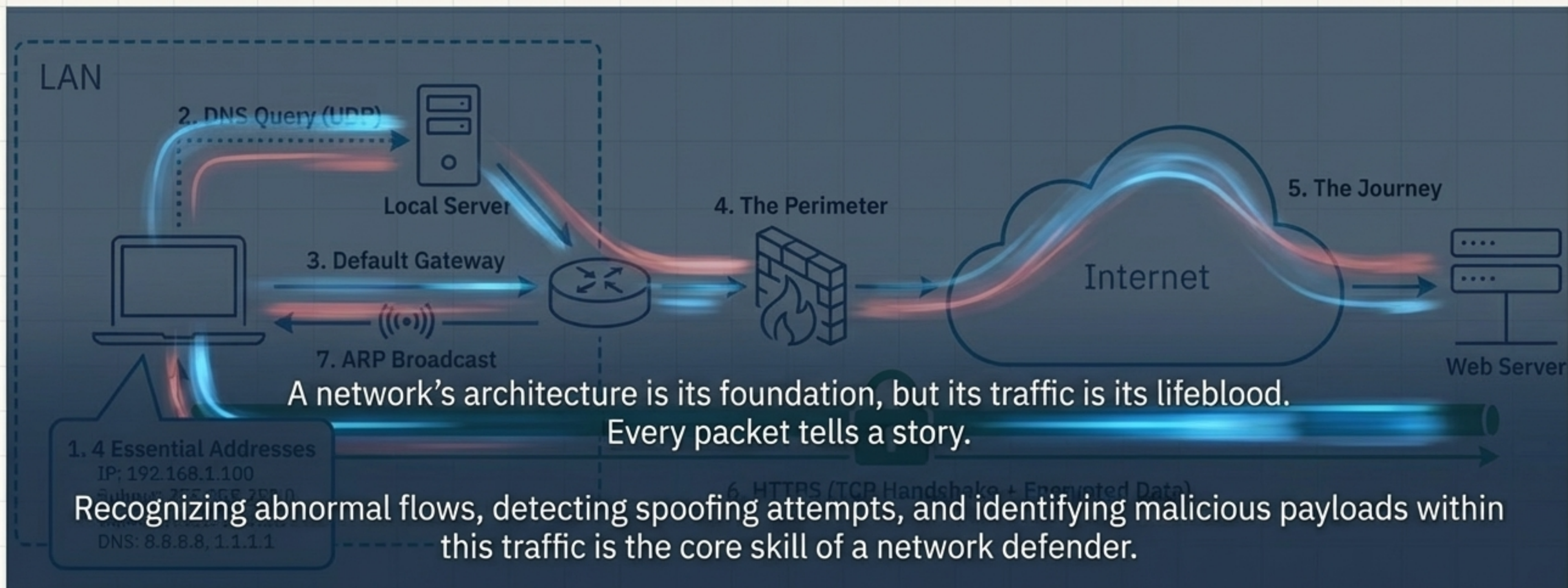
Instructor Note: Understanding these fundamental vulnerabilities is the prerequisite for effectively using network devices and traffic analysis tools for defense.

The Complete Blueprint: Visualizing a Secure Connection



From a local ARP request to a global HTTPS session, every component has a role. A defender must be able to see and understand the entire chain.

You Understand the Blueprint. Now, Watch What Moves Across It.



Having mastered the blueprint, our next step is to learn the tools and techniques of Traffic Analysis and Intrusion Detection.